

## 2 Logic

### 2.1 Statements

**Decide whether or not the following are statements. In the case of a statement, say if it is true or false, if possible.**

- 2. Every even integer is a real number.  $\rightarrow$  Statement, true.
- 4. Sets  $\mathbb{Z}$  and  $\mathbb{N}$ .  $\rightarrow$  Not a statement.
- 6. Some sets are finite.  $\rightarrow$  Statement, true.
- 8.  $\mathbb{N} \notin \mathcal{P}(\mathbb{N}) \rightarrow$  Statement, false.
- 10.  $(\mathbb{R} \times \mathbb{N}) \cap (\mathbb{N} \times \mathbb{R}) = \mathbb{N} \times \mathbb{N}$ .  $\rightarrow$  Statement, true.
- 12. If the integer  $x$  is a multiple of 7, then it is divisible by 7.  $\rightarrow$  Statement, true.
- 14. Call me Ishmael.  $\rightarrow$  Not a statement.

### 2.2 And, Or, Not

**Express each statement or open sentence in one of the forms  $P \wedge Q$ ,  $P \vee Q$ , or  $\neg P$ . Be sure to also state exactly what statements  $P$  and  $Q$  stand for.**

- 2. The matrix  $A$  is not invertible.  $\rightarrow \neg P$ , where  $P$ : Matrix  $A$  is invertible.
- 4.  $x < y$ .  $\rightarrow \neg P$ , where  $P$ :  $x \geq y$
- 6. There is a quiz scheduled for Wednesday or Friday.  $\rightarrow P \vee Q$ , where  $P$ : A quiz is scheduled for Wednesday, and  $Q$ : A quiz is scheduled for Friday.
- 8. At least one of the numbers  $x$  and  $y$  equals 0.  $\rightarrow P \vee Q$ , where  $P$ :  $x = 0$ , and  $Q$ :  $y = 0$ .

10.  $x \in A \cup B. \rightarrow P \vee Q$ , where  $P$ :  $x \in A$ , and  $Q$ :  $x \in B$ .
12. Happy families are all alike, but each unhappy family is unhappy in its own way. (*Leo Tolstoy, Anna Karenina*)  $\rightarrow P \wedge Q$ , where  $P$ : Happy families are all alike, and  $Q$ : Each unhappy family is unhappy in its own way.
14. A man should look for what is, and not for what he thinks should be. (*Albert Einstein*)  $\rightarrow P \wedge \neg Q$ , where  $P$ : Man should look for what is, and  $Q$ : Man should look for what he thinks should be.